

ReqGen 1.1.0

GLOBAL STABILISATION & IMPLEMENTATION ROADMAP

Authoritative implementation handoff based on the current laptop RAR baseline audit. Purpose: consolidate architecture, restore complete administrative control, remove legacy modules, standardise responsive UI, enforce role-based access, and guarantee live-data integrity before final production submission.

LOCKED ARCHITECTURAL DECISION: The standalone HR module is to be removed completely. HR and Registrar remain canonical user roles with authorised workflow responsibilities; they are not standalone application modules.

Baseline Executive Summary

Area	Current RAR finding	Target state
Finance	8 consolidated workspaces	Preserve and harden
Payment Vouchers	Centre + PV Settings	Preserve and harden
Registry	Centre + Archive	Preserve and harden
Reports / Analytics	Admin + Auditor only	Preserve
Workflow	Standalone page still present	Remove UI; preserve engine
Admin Departments	CRUD split/hidden	One canonical full CRUD workspace
Subheads	CRUD exists under Finance	Expose canonical Admin entry
User deletion	Missing	Privileged server-side deletion
Responsive UI	System-wide inconsistencies	Shared responsive design system
Legacy HR/Staff/Executive	Still physically present	Dependency-audit then remove/redirect

PHASE 0 — Baseline Freeze, Inventory & Recovery Control

Objective. Establish the current laptop RAR as the single authoritative baseline and prevent older ZIP packages or legacy patches from reintroducing regressions.

Implementation scope

- Preserve the original RAR unchanged and generate a file manifest with hashes for all source, configuration, migration, script and public assets.
- Record framework/runtime versions, package dependencies, Supabase configuration expectations, environment-variable names and deployment commands.
- Inventory every physical route and classify it as ACTIVE, CONTEXTUAL, REDIRECT, LEGACY or REMOVE.
- Inventory every sidebar item, global-search entry, breadcrumb target, shortcut, button destination and role gate against the route register.
- Create a formal baseline report so every later phase can be diffed against the same source.
- Do not mix files from the earlier ZIP into the RAR baseline unless a specific reviewed change is intentionally ported.

Acceptance gate

- Authoritative baseline identified and backed up.
- Complete route/file manifest exists.
- No unidentified page or navigation target remains.
- All subsequent work branches from the same baseline.

PHASE 1 — Core Architecture, Navigation & Route Reconciliation

Objective. Repair the application skeleton so navigation, collapsibles, search, breadcrumbs, route protection and contextual pages behave consistently.

Implementation scope

- Create one canonical route/navigation registry consumed by desktop sidebar, mobile navigation, global search, breadcrumbs and active-module detection.
- Fix collapsible behaviour globally: clicking an expanded navigation group must collapse it; clicking a closed group must expand it predictably.
- Remove standalone Workflow from visible navigation and global search while preserving workflow stages, routing, current-owner logic, RPCs and history.
- Redirect the old /workflow route to the appropriate Audit Centre request/workflow-trace view.
- Reconcile compatibility routes so old URLs either redirect safely or remain contextual without appearing as duplicate modules.
- Remove stale Command Centre/legacy Executive exposure from normal navigation.
- Repair stale audit scripts so they enforce the current consolidated architecture rather than superseded August navigation.
- Restore the missing Section 4 audit command/script or replace it with the correct current equivalent.

Acceptance gate

- No broken internal links.

- No duplicate route policies.
- Every visible navigation item has a valid authorised destination.
- Every contextual/hidden page has a deliberate parent or redirect.
- Collapsibles work consistently on desktop and mobile.
- Workflow engine remains operational although the standalone Workflow module is gone.

PHASE 2 — Super Admin Control Recovery

Objective. Give Admin/Super Admin complete, discoverable control of Departments, Subheads and users without secret URLs or split management screens.

Implementation scope

- Merge /admin/departments routing controls with the richer /finance/departments CRUD into one canonical Admin Department Management workspace.
- Department Management must support Add, Edit/Rename, Activate/Deactivate and safe Delete, plus HOD and Director assignment.
- Show linked users, requests and subheads so destructive actions can be blocked when dependencies exist.
- Ensure Welfare and any future department can be created from the UI and immediately becomes available to all live department selectors.
- Expose a clear Admin → Subheads & Budget Structure entry to the canonical subhead management workspace; avoid duplicate CRUD implementations.
- Subhead management must support Add, Edit, allocation, department linkage, IET account linkage, Activate/Deactivate and safe Delete.
- Implement Admin Delete User through a protected server-side endpoint using the Supabase service-role credential; never expose that credential to browser code.
- Delete Auth identity and removable role/profile relationships while preserving immutable historical requests, approvals, finance, PV and audit evidence.
- Prevent accidental self-deletion and require explicit confirmation for destructive administration actions.
- Keep password, username/email and MFA self-service in Profile/Security rather than duplicating recovery controls in Admin.

Acceptance gate

- Admin can add Welfare without SQL/manual database intervention.
- Admin can rename/edit departments and manage routing officers.
- Admin can create/edit subheads from a clearly discoverable location.
- All dropdowns receive newly created departments/subheads from authoritative Supabase data.
- Admin user deletion works server-side and historical records remain intelligible.

PHASE 3 — Canonical RBAC & Direct-URL Security Hardening

Objective. Make role permissions deterministic across navigation visibility, page access, API actions and direct URL entry.

Implementation scope

- Use canonical roles: Staff, Director, DG, Account Officer, Auditor, Admin, HR, Registrar, General Secretary and Dean Admin.
- HR and Registrar are roles only. Remove the standalone HR module and do not recreate a separate Registrar module unless a role-specific operational workspace is explicitly required.
- Staff/Director/DG/HR/General Secretary/Dean Admin receive only their authorised core modules; Registrar additionally receives authorised Registry operations.
- Account Officer receives Finance and Payment Voucher operational access as approved.
- Auditor receives authorised Finance/PV oversight, Reports, Audit Centre and Registry oversight.
- Admin receives full authorised administration access.
- Reports & Executive Analytics remain Admin + Auditor only; DG has no system access to those modules.
- PV signing/countersigning permissions must not automatically grant Finance access.
- Test every route against every canonical role, including direct URL entry, not only sidebar hiding.
- Server/API/RPC-sensitive actions must independently validate authorisation.

Acceptance gate

- A route × role access matrix is generated and passes.
- No unclassified sensitive route.
- No role can gain access by typing a hidden URL.
- Navigation visibility and actual route/API permission agree.

PHASE 4 — Global Responsive UI & Design-System Standardisation

Objective. Eliminate overlapping text, cramped controls, inconsistent spacing and desktop-only layouts by enforcing reusable responsive primitives.

Implementation scope

- Create/reconcile shared primitives for Page, Header, Toolbar, Filter Bar, KPI Grid, Card, Data Table, Form Grid, Workspace, Empty State and confirmation surfaces.
- Preserve the adopted ReqGen shell and 13px navigation/collapsible typography.
- Consolidate globals.css: remove contradictory generations of overrides and move stable values into a single token/layout layer.
- Standardise spacing, card padding, section gaps, field spacing, button heights, table density, border radii and content widths.
- Define predictable mobile, tablet, laptop and wide-desktop breakpoints.
- KPI and form grids must reflow automatically instead of forcing excessive columns.
- Header actions and filter controls must wrap without covering titles or other controls.
- Use horizontal scrolling only for genuinely wide data tables; keep primary identifiers/actions visible where practical.
- Audit fixed widths, min-width tables, 420px popovers, 1200px/1150px tables and large multi-column Admin grids.
- Ensure no text, button, badge, card, chart, dropdown or modal overlaps another component.

- Apply the same responsive behaviour to page-specific styles rather than relying only on global overrides.
- Maintain sufficient whitespace even when scrolling is required; fitting everything above the fold is not a requirement.

Acceptance gate

- Responsive audit warnings reduced to reviewed/justified exceptions only.
- No component overlap at supported breakpoints.
- No clipped action buttons or unreadable labels.
- All major workspaces usable on mobile, tablet and desktop.
- Approved shell remains visually consistent across modules.

PHASE 5 — Live Data Integrity & Single-Source Calculations

Objective. Ensure every KPI, chart, table, dropdown and report is driven by authoritative Supabase data and consistent domain calculations.

Implementation scope

- Create/reconcile shared data-access modules for departments, subheads, requests, finance, vouchers, registry, analytics and audit data.
- Use one canonical subhead balance formula everywhere: Approved Allocation – Reserved – Expenditure.
- Reconcile the business rule for DG-approved funded requests with finance posting logic before changing database posting, to prevent double expenditure.
- Remove fabricated fallback values, decorative production statistics and silent data substitutions.
- Do not truncate departments, subheads or IET accounts unless the UI explicitly labels a Top-N view.
- All charts must disclose/represent the same filtered dataset used by related KPI totals where logically applicable.
- All department/subhead/account selectors must query live authoritative tables and respect active/inactive status rules.
- Reports, Finance, PV and Analytics must reuse the same financial calculations.
- Add loading, empty, error and unavailable-data states instead of inventing values.
- Review real-time subscriptions only where they add operational value and ensure refresh paths are deterministic.

Acceptance gate

- Data-source audit passes for every production KPI/chart/dropdown/table.
- Same business metric produces the same value across modules.
- No misleading or synthetic production data.
- All IET accounts and relevant departments/subheads are represented correctly.

PHASE 6 — Approved Mockup & Admin Page Reconciliation

Objective. Fish out pages that diverged from approved designs and bring them back into the adopted shell without sacrificing live functionality.

Implementation scope

- Audit every Admin route and compare it with the approved mockup/design intent.
- Classify each Admin page as KEEP, MERGE, REDIRECT, REMOVE or REDESIGN.
- Preserve already approved/locked Dashboard, Requests, Approvals, Finance, PV, Registry and Reports structures unless a defect requires a targeted fix.
- Correct page-specific spacing, typography, toolbar placement, form density, card layout and table presentation.
- Do not duplicate pages merely to match an old mockup; reconcile the mockup with the canonical route architecture.
- Where no trustworthy approved mockup exists, create a new mockup for approval before major visual redesign.
- Maintain live data and authorised actions while matching the adopted visual language.

Acceptance gate

- Every active Admin page has an approved visual target.
- No active page uses an unexplained legacy layout.
- Similar workspaces share the same structural components.
- No redesign replaces live data with placeholders.

PHASE 7 — Legacy Module Removal & Repository Simplification

Objective. Remove obsolete physical code only after dependency analysis proves it is safe, reducing maintenance burden and preventing accidental re-exposure.

Implementation scope

- Remove the standalone HR module completely after confirming required workflow logic has been relocated or is role-based.
- Preserve HR as a canonical role only, with authorised responsibilities integrated into Requests/Approvals or other approved workspaces.
- Preserve Registrar as a canonical role, including approved Registry operational access; do not treat Registrar as a standalone module.
- Dependency-audit and remove/redirect obsolete Staff module pages.
- Dependency-audit and remove/redirect obsolete Executive/Command Centre pages.
- Remove retired Workflow UI after redirect and monitoring migration are verified.
- Review obsolete Finance/PV/Registry compatibility pages and keep only redirects/contextual pages still required.
- Review /output, test, docs/about and other utility routes for production necessity.
- Remove orphan components, stale CSS, unused navigation entries and obsolete audit expectations.
- Keep migrations/history documentation needed for traceability even when obsolete runtime UI is removed.

Acceptance gate

- No obsolete module is exposed in navigation/search.
- No deleted route is referenced by live code.
- HR and Registrar roles continue to function after HR module removal.

- Repository route count is reduced to deliberate active/contextual/redirect pages.
- Build and route audits pass after cleanup.

Cross-Phase Production Gates

- Run lint with zero unresolved production errors.
- Run TypeScript type checking with zero errors.
- Run Next.js production build successfully.
- Run route/navigation integrity audit with zero broken internal targets.
- Run RBAC/direct-URL matrix audit.
- Run responsive-layout audit and manually review justified exceptions.
- Run live-data/data-contract audit for Finance, PV, Registry, Reports, Analytics and Admin selectors.
- Run Section-specific audits after updating stale expectations.
- Perform browser QA at representative mobile, tablet, laptop and wide-desktop widths.
- Verify Supabase migrations/RPC expectations against the deployed environment before release.
- Produce a final deployment handoff with changed files, migrations, environment requirements, rollback notes and acceptance results.

Locked Functional Decisions

Decision	Locked requirement
HR	Standalone HR module removed completely; HR remains a role.
Registrar	Registrar remains a role; authorised Registry access is role-based.
Workflow	Standalone UI removed; backend workflow engine/history preserved.
Reports	Admin + Auditor only; DG receives hard-copy reporting outside system.
Departments	Canonical Admin CRUD + HOD/Director routing; Welfare can be added dynamically.
Subheads	Canonical live CRUD exposed clearly to Admin; shared with authorised Finance operations.
Users	Admin can securely delete user/Auth identity while preserving immutable historical evidence.
Navigation typography	13px baseline for main and collapsible navigation.
Financial balance	Approved Allocation – Reserved – Expenditure.
Production data	No fabricated/fallback production values or misleading charts.

Recommended Implementation Order

0 → 1 → 2 → 3 → 4 → 5 → 6 → 7. Architecture and Admin operational controls come before broad visual polishing. This prevents the team from polishing pages that will later be merged, redirected or removed.

New-Chat Handoff Instruction

In the new project chat, attach the current laptop RAR and this roadmap. Treat the RAR as the authoritative baseline. Begin at Phase 0 and maintain a phase-by-phase change register. Do not import the older ZIP wholesale. For each phase: inspect → implement → run targeted audits → run lint/type/build gates → provide patch/full package as appropriate → obtain approval → lock the phase.

The objective is not a cosmetic redesign alone. The completion criterion is a coherent, secure, responsive and auditable ReqGen system in which every active page is discoverable when

authorised, every hidden/contextual page is deliberate, every role is correctly enforced, and every operational number is traceable to authoritative data.